

Mohammed Omar Elsiddieg Abdallah

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Bio

Biomedical Data Scientist, interested in integrating AI , NLP and Knowledge Graphs in healthcare and biomedical research. Expert in scalable genomics analyses, and workflow management.

Academic Qualifications

Ph.D. (Molecular Biology)

May 2019

University of Khartoum, Khartoum, Sudan

M. Sc Molecular Medicine

Jul 2012

University of Khartoum, Khartoum, Sudan

Bachelor of Medicine and Surgery

Apr 2005

University of Khartoum, Khartoum, Sudan

Licensure

Sudan Medical Council Registration Number: 21533. Division: Full

Most recent position

Data Scientist

January 2024

Data, Analytics, Reporting & Evaluation (**DARE**) Provincial Health Services Authority (**PHSA**)
Vancouver, BC, Canada

Managing clinical databases (CDM), Posit servers, supporting R and Python analyses and overseeing AI and NLP projects for healthcare.

Notable Projects

Silent Genome Indigenous Background Variation Library (IBVL)

African Genetics Variation Database (AGVD)

The Sudanese Genetic Variation Portal (SGVP)

Research Interests

Deep Learning and AI for biomedical research

Scalable Genomics workflows

Knowledge Graphs for Biological Data Integration

Recent Work and Research Experience

Postdoctoral Fellow

May 2021-Dec 2023

Centre for Molecular Medicine and Therapeutics, University of British Columbia

Duties: Designing genomic pipelines for the Silent Genome Project including developing and maintaining Nextflow pipelines for population-level WGS analysis. Deep Learning with multi-omics knowledge graphs.

Postdoctoral Fellow (Human Genetics)

Feb 2019-April 2021

H3AbioNet Pan African Bioinformatics Network for the Human Heredity and Health in Africa (H3Africa) consortium: African Genetic Variation Database technical team. I oversaw OpenCGA deployment and training & MRC Genomic and Precision Medicine Research Unit, Institute of Infectious Diseases and Molecular Medicine & University of Cape Town, South Africa

Head of the bioinformatics and NGS data analysis team

Mar 2015-May 2018

Institute of Endemic Diseases, University of

Khartoum

Duties: Include analysis of Exome Data for rare genetics disorders including familial cancers as well as teaching and instructing Bioinformatics workshops

Lecturer

Apr 2017 – Jan 2019

Faculty of dentistry, university of Khartoum

Department of basic medical sciences

Duties: Teaching molecular genetics to undergraduate students

Visiting Researcher

Apr 2016 - -Jul 2016

Cambridge University, UK

Duties: Joined the OpenCGA team, and developed R and Python clients to the OpenCB web services

Researcher (Short-term Fellowship)

Mar-April 2013

International Center of Genetic Engineering and Biotechnology Trieste, Italy

Duties: Include conducting plasmid DNA extraction, cell culture, transfection, and using immunofluorescence techniques to study the role of Epstein Bar Virus in Breast cancer

Teaching Assistant

Sep 2008 –Aug 2012

Department of Microbiology, Faculty of Medicine, International University of Africa, Khartoum, Sudan.

Duties: include Preparing and Instructing Medical microbiology lab

Medical Officer

Jun 2007 – Jun 2008

Ministry of Health Khartoum, Sudan.

Duties: include participating in inpatient and outpatient clinic activities, attending clinical rounds, tutorials, and various clinical and training activities.

House Officer

Dec 2005-May 2007

Duties: include participating in inpatient and outpatient clinic activities, attending clinical rounds, tutorials, and various clinical and training activities

Academic and Computer Skills

AI: RAG, PEFT LLM Finetuning, Structured data extraction

Advanced Computer Skills in Windows, Mac, and **Linux** Platforms Strong Experience with **Linux Servers, Cloud Computing training in Azure and GCP**

Extensive Experience in **High-Performance Computing (Compute Canada, UCT HPC, BCCHR GPCC)**

Advanced Experience in R statistical Language (**R developer, Posit Administrator**)

Advanced skills in **Biomedical NLP analysis and knowledge Extraction**

Strong experience with **Apache Spark and SQL (Spark SQL, SQL server, Postgres)**

Advanced Experience with scientific Python (**Jupyter, NumPy, Pandas, Scikit learn, etc**)

Strong experience in interactive web-based analytical dashboards in both R, (**Shiny**) and **Python (Streamlit)**

Strong experience with container technology and reproducible research (**docker, git, markdown**)

Strong experience in graph databases (**Neo4J and heterogeneous biomedical Knowledge Graphs**)

End-to-end genomics expertise in genetic variants analysis in NGS, GWAS, and cancer genomics.

Fluent in **Bioinformatics Pipelines, Nextflow.**

Advanced Skills in Microsoft Office Software (Word, Excel, and PowerPoint)

Native fluency in Arabic and advanced fluency in English

Select Open-Source Software Contributions

The variant catalogue Pipeline is a **Nextflow** workflow designed to generate variant catalogues, a list of variants and their frequencies in a population, from whole genome sequences

https://github.com/wassermanlab/Variant_catalogue_pipeline

Python Learning Project: an outreach program to teach Python programming to Indigenous high-school students using Raspberry Pi and Jupyter notebooks.

<https://github.com/wassermanlab/plp>

CellbaseR A **Bioconductor R** package for querying the CellBase web services.

<https://www.bioconductor.org/packages/release/bioc/html/cellbaseR.html>

The Sudanese Genetic Variation Portal, A Shiny App to explore Sudanese Genetic Variation Data with extensive clinical and pharmacological annotations

<https://github.com/melsiddieg/SGP>

Mycetoma Data Explorer, A shiny App to explore Mycetoma historical data in Sudan

https://melsiddieg.shinyapps.io/mycetoma_app/

Map of Mycetoma distribution in Sudan

https://github.com/melsiddieg/Mycetoma_Map

Additional Courses and Training:

Microsoft ESI DP-601 Implementing a Lakehouse with Microsoft Fabric (Dec 2024)

DP-602 Implementing a Data Warehouse with Microsoft Fabric (Dec 2024)

Posit Admin Training (September 2024)

Generative AI with Large Language Models (Coursera July 20 2023)

edX Honor Code Certificate for Explore Statistics with R (July 2015)

edX Verified Certificate for Case study: DNA methylation data analysis (June 2015)

edX Verified Certificate for Case Study: Variant Discovery and Genotyping (June 2016)

edX Honor Code Certificate for Introduction to Bioconductor ((May 2015)

edX Honor Code Certificate for Statistics and R for the Life Sciences (May 2015)

edX Honor Code Certificate for Data Analysis for Genomics (July 2014)

Khartoum winter School of Bioinformatics and Statistical Genetics,

Khartoum, Sudan (January 2013)

Parallel computing in Biomedical Research

Khartoum, Sudan (December 2012)

International Workshop on Molecular Immunology and Drug Discovery

Khartoum, Sudan (July 2010)

Publications

Lost in translation: the pitfalls of Ensembl gene annotations between human genome assemblies and their impact on diagnostics *Mohammed O.E Abdallah*, Mahmoud Koko & Raj Ramesar *Expert Review of Molecular Diagnostics* (July 2023) <https://doi.org/10.1080/14737159.2023.2237411>

Downregulation of MGMT expression by targeted editing of DNA methylation enhances temozolomide sensitivity in glioblastoma Xinyu Han a, *Mohammed O.E. Abdallah* b, Peter Breuer a, Fabian Stahl a,c, Yousuf Bakhit a,d, Anna-Laura Potthoff e, Barbara E.F. Pregler e, Matthias Schneider e, Andreas Waha f, Ullrich Wüllner a,c,* , Bernd O. Evert *Neoplasia* (August 2023) <https://doi.org/10.1016/j.neo.2023.100929>

EBV Associated Breast Cancer Whole Methylome Analysis Reveals Viral and Developmental Enriched Pathways *Mohammad O. E. Abdallah*, Ubai K. Algizouli, Maram A. Suliman, Rawya A. Abdulrahman, Mahmoud Koko, Ghimja Fessahaye, Jamal H. Shakir, Ahmed H. Fahal, Ahmed M. Elhassan, Muntaser E. Ibrahim and Hiba S. Mohamed *Frontiers in oncology* (August 2018) <https://doi.org/10.3389/fonc.2018.00316>

Age, absolute CD4 count, and CD4 percentage in relation to HPV infection and the stage of cervical disease in HIV-1-positive women. Ramadhani Chambuso Raj Ramesar, Evelyn Kaambo , Alltalents T Murahwa , *Mohammed O E Abdallah* , Michelle De Sousa , Lynette Denny , Anna-Lise Williamson Clive M Gray *Medicine* 2020 <https://doi.org/10.1097/md.00000000000019273>

Challenges imposed by minor reference alleles on the identification and reporting of clinical variants from exome data Mahmoud Koko *Mohammed O. E. Abdallah*¹, Mutaz Amin and Muntaser Ibrahim *BMC Genomics* (2018) 19:46 <https://doi.org/10.1186/s12864-018-4433-3>

Exome sequencing of a colorectal cancer family reveal shared mutation pattern and predisposition circuitry along tumour pathways Suleiman Suleiman H, Koko Mahmoud E, Nasir Wafaa H, Elfateh Ommnyiah, Elgizouli Ubai K, *Abdallah Mohammed O E*, Alfrouk Khalid O, Hussain Ayman, Faisal

Shima, Ibrahim Fathelrahman M A, Romano Maurizio, Sultan Ali, Banks Lawrence, Newport Melanie, Baralle Francesco, Elhassan Ahmed M, Mohamed Hiba S, Ibrahim Muntaser E. *Frontiers in Genetics*. 6, 2015. <https://doi.org/10.3389/fgene.2015.00288>

The variant catalogue pipeline: A workflow to generate a background variant library from Whole Genome Sequences Solenne Correard, *Mohammed OE Abdallah*, Brittany Hewitson, Wyeth W. Wasserman <https://www.biorxiv.org/content/10.1101/2022.10.03.508010v2> (in review)

Autosomal recessive Leber Congenital Amaurosis type 4 resulting from AIPL1 truncating variant c.487C>T: support of pathogenicity *Mohammed O. E. Abdallah*, Mahmoud E. Koko1, Shima Faisal, Melanie J. Newport , Muntaser E. Ibrahim <https://www.biorxiv.org/content/10.1101/290650> (preprint)

Unpublished Works

CellbaseR, a scalable R package for retrieving biological data and performing variant annotation (In preparation)

Whole Exome Analysis of Adult Diffuse Glioma Patients Reveals Patterns of Somatic Mutations (In preparation)

Genetic architecture of schizophrenia in the South African Xhosa population (in preparation)

Invited talks, Workshops, and Community Work

The Sudanese Genetic Variation Portal, enabling personalized medicine in understudied populations G2MC conference 27th -30th November 2018 **Cape Town, South Africa**

The impact of open source data and NGS revolution on population genetics
“Health, Society, Biology and the History of East Africans” Regional Meeting in Sudan
February 27-March 2 2017

Introduction to Bioinformatics (lecture series)
Biomedical Informatics Society, Faculty of Medicine, University of Khartoum July 2016-2018

Introduction to R and Bioconductor
Center for Bioinformatics and Systems Biology (CBSB) Khartoum Node for H3AbioNet
Faculty of Science, University of Khartoum January. 10th - 13th 2016

Introduction to R and NGS
Khartoum Winter School of Bioinformatics and Statistical Genetics 2014-2017

Awards Received

Research Development Fund	August 2015
Wellcome Trust Brighton and Sussex Centre for Global Health Research	

Arturo Falaschi ICGEB Short-term Fellowship	March-April 2013
ICGEB, Italy	

Miscellanies

International English Language Testing System (IELTS) score: **8.0**

References

Available upon request